

EECS445 Project 1 Report

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Abstract

The project 1 of EECS445 (WN25) is **Predicting Survival in the ICU**. The yellow parts (required written answers) are presented in this report.

(This part is named as "abstract" to align the section number with P1 Spec.)

1 Feature Extraction

- (a) i)
 - **Question:** 1. How might such a representation impose unintended meaning on the feature? 2. What are the implications of that when using a linear classifier? 3. How else might you represent this variable (as possibly more than one feature)?
 - **Answer:**
 1. Such a representation might impose unintended meaning on the **ordering** of the 4 different ICU Types. Moreover, it will also imply 3 is closer to 1 and 2 than to 4, which implies an unintended **"distance"** meaning between different types of ICU. In other words, it may be interpreted as a **continuous numerical value** other than a categorical discrete value as intended.
 2. When using a linear classifier, this will imply a **continuous relationship between 4 categories**. The decision boundary will be misleading since the classifier won't treat 4 types of ICU types **independently**. The **coefficient** before it is also **hard to interpret**.
 3. I might represent it as **4 binary variables** as *is_Coronary_Care_Unit*, *is_Cardiac_Surgery_Recovery_Unit*, *is_Medical_ICU*, *is_Surgical_ICU*.
- (a) ii)
 - **Question:** What limitations are associated with this representation? What other summary statistics could be useful?

- **Answer:** The limitations include only considering extreme conditions (also *min* is more informative for some attributes) **without monitoring the process**. Some other statistics such as **mean value, minimum, variance** could also be useful for some attributes.
- (b)
 - **Question:** What assumptions does this imputation approach make? How else might you handle missing data?
 - **Answer:** The approach assumes every missing data is **randomly distributed**, and is **independent from other features**. Thus, this approach could choose mean as the representative statistics. Some other methods to handle the missing data include **choosing median as the representative statistics**.
- (c)
 - **Question:** Without feature normalization, regularization can affect some features more than others. Explain how.
 - **Answer:** From my view, regularization will present equal penalty (e.g. in ridge regression, it's $\lambda ||\bar{\theta}||_2$.) to each features. To compensate for the penalty, the algorithm will prefer to shrink the coefficients of the large-scale data than small-scale ones. In a word, **regularization will shrink the coefficients before large-scale data more severely** than small-scale ones without normalization.
- (d)
 - **Question:** Report the name, range for each of the 40 features in the training set mean value, and interquartile X_{train} in a table.
 - **Answer:** See Table 1.

2 Hyperparameter and Model Selection

2.1 Hyperparameter Selection for a Regularized Logistic Regression

- (a)
 - **Question:** In your write-up, briefly describe why it might be beneficial to maintain class proportions across folds.
 - **Answer:** Maintaining class proportions across folds will **reduce bias** and provide more **consistent** cross-validation results.
- (b)

Feature Name	Mean	IQR
Age	0.65	0.34
Gender	0.55	1.00
Height	0.39	0.00
ICUType	0.58	0.67
Weight	0.23	0.13
max_ALP	0.04	0.01
max_ALT	0.02	0.02
max_AST	0.02	0.01
max_Albumin	0.50	0.00
max_BUN	0.14	0.11
max_Bilirubin	0.05	0.03
max_Cholesterol	0.36	0.00
max_Creatinine	0.07	0.04
max_DiasABP	0.31	0.04
max_FiO2	0.76	0.24
max_GCS	0.88	0.17
max_Glucose	0.10	0.06
max_HCO3	0.36	0.13
max_HCT	0.30	0.16
max_HR	0.22	0.11
max_K	0.10	0.03
max_Lactate	0.09	0.03
max_MAP	0.39	0.05
max_Mg	0.13	0.05
max_NIDiasABP	0.44	0.12
max_NIMAP	0.49	0.11
max_NISysABP	0.53	0.11
max_Na	0.38	0.09
max_PaCO2	0.38	0.10
max_PaO2	0.48	0.30
max_Platelets	0.20	0.11
max_RespRate	0.16	0.00
max_SaO2	0.91	0.01
max_SysABP	0.53	0.07
max_Temp	0.35	0.14
max_TroponinI	0.18	0.00
max_TroponinT	0.05	0.00
max_Urine	0.11	0.08
max_WBC	0.10	0.05
max_pH	0.00	0.00

Table 1: Summary of Feature Means and Interquartile Ranges for X_{train}

- **Question:** In your write-up, describe the role of C . What do we expect to happen as we increase the value the user of C ?
 - **Answer:** C is used to **adjust the trade-off between regularization and model complexity**. As C increases, the model becomes **less regularized and more complicated**.
- (c)
 - **Question:** Report your findings in tabular format with four columns: names of the performance measures, along with corresponding values of C , regularization penalty and the mean, min and max cross-validation performance.
 - **Answer:** See Figure 1.

Performance Measures	C	Penalty	Mean (Min, Max) CV Performance
Accuracy	1	λ_2	0.8637 (0.8531, 0.8844)
Precision	1	λ_2	0.5740 (0.4000, 0.9000)
F1-Score	100	λ_2	0.2926 (0.2295, 0.3750)
AUROC	1	λ_1	0.8002 (0.7530, 0.8592)
Average Precision	1	λ_1	0.4429 (0.3751, 0.6017)
Sensitivity	100	λ_2	0.2071 (0.1556, 0.2667)
Specificity	0.01	λ_1	1.0000 (1.0000, 1.0000)

Table 1: Cross-validation performance with different hyperparameters

Figure 1: Table of 2c

- **Question:** Also, in your write-up, describe how the 5-fold CV performance varies with C for specific metrics. Optimize which performance measure? Explain.
 - **Answer:** The accuracy increases as C increases. The precision increases first, then decreases as C increases. The sensitivity decreases as C increases. The f1-score increases first, then decreases as C increases. The AUROC increases as C increases. The specificity drops as it increases.
I'll choose to optimize the sensitivity to reduce the false negatives (i.e. misidentifying a dying patient to be negative). The reason is that misidentifying a dying patient as a survivor will make this patient not enough cared, leading to many deaths.
- (d)
 - **Question:** Report the performance of this model on the test data X_{test}, y_{test} for each metric in Table 2.
 - **Answer:** See Figure 2.

$C = 1$ penalty = l_1	
Performance Measures	Test Performance
Accuracy	0.8650
Precision	0.6000
F1-score	0.2500
AUROC	0.8139
Average Precision	0.4956
Sensitivity	0.1579
Specificity	0.9825

Table 2: Test Performance for Logistic Regression with C and penalty from 2.1.(c)

Figure 2: Table of 2d

- (e)
 - **Question:** In your write-up, include the produced plot and describe any trends you observe.
 - **Answer:** See Figure 3. The trend is that for "l2-norm", the norm of θ keeps around 40; for "l1-norm", the norm of θ increases from 0 to 40 as C goes up.
- (f)
 - **Question:** Report both the coefficients and the corresponding features in Table 2f.
 - **Answer:** See Figure 4.

3 Asymmetric Cost Functions and Class Imbalance

3.1 Arbitrary class weights

- (a)
 - **Question:** If W_p is much greater than W_n , what does this mean in terms of classifying positive and negative points? Describe how this modification may affect the solution.
 - **Answer:** If W_p is much greater, then the classifier tends to **less misclassify the positive points** since more penalty is distributed on the positive points. This modification will successfully deal with the class imbalance in our case since we have much smaller proportion of positive samples.

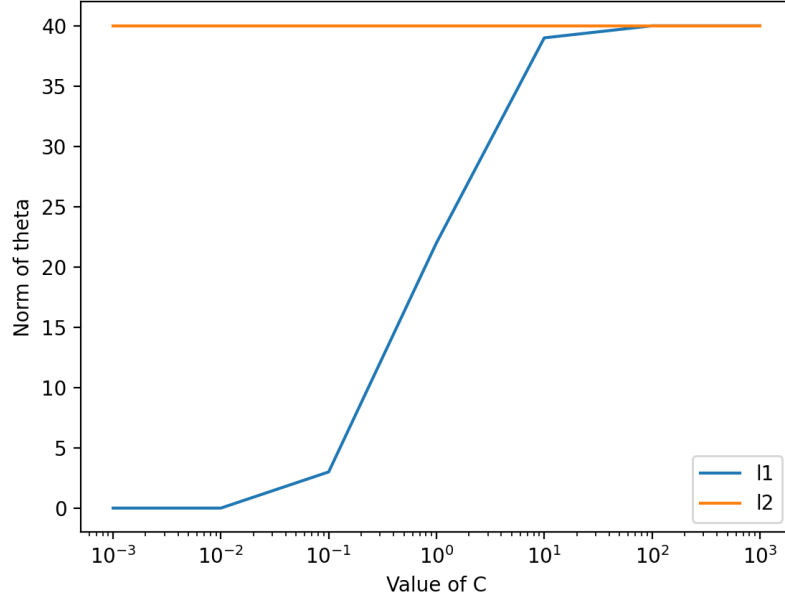


Figure 3: L_0 norm.

- (b)
 - **Question:** Report the performance for the modified linear model on the test data X_{test} , y_{test} for each metric in Table 3.1(b).
 - **Answer:** See Table 5.
- (c)
 - **Question:** Which performance measures are affected the most by the new class weights? Which performance measures are less affected? Why do you suspect this is the case?
 - **Answer:** **Sensitivity, Specificity, Precision, and Accuracy** are most affected ones; **F1-score, average precision, and AUROC** are less affected.

The reason behind is that increasing positive sample loss' coefficients will let more penalty on positive samples, less on negative samples, making it easier to misclassify negative samples. That will decrease accuracy (negative is majority), precision, and specificity. On the other hand, the sensitivity will greatly increase due to the increased penalty. For other performance measures, they are based on balanced

Positive Coefficient	Feature Name	Negative Coefficient	Feature Name
3.610903	max_Lactate	-2.804441	max-GCS
3.326896	max-BUN	-2.312801	max-Urine
2.930581	max-Bilirubin	-1.147259	max-Temp
2.154956	Age	-0.830326	max-PaO ₂

Table 3: Most predictive features

Figure 4: Table of 2f.

performance of mis-classification of positive and negative samples, so they don't change a lot.

3.2 Choosing appropriate class weights

- (a)
 - **Question:** Find an appropriate setting for the class weights.
 - **Answer:** My chosen setting is 1:4.
 - **Question:** Report your choice for W_n and W_p and describe your strategy for choosing these parameters.
 - **Answer:** I chose $W_n = 1$, $W_p = 4$. I chose it so that it will optimize the mean performance with metric **AUROC** (at the range of 1:1 to 1:50, with W_n set as 1).
- (b)
 - **Question:** Report the performance of the test set, for each metric in Table 3.2b.
 - **Answer:** See Figure 7.

3.3 The ROC curve

- (a)
 - **Question:** First, provide a plot of the ROC curves with labeled axes, using $C=1.0$, for both two cases.
 - **Answer:** See Figure 3.3a.
- (b)
 - **Question:** In your write-up, briefly propose an alternate method for fixing a classifier.
 - **Answer:** The alternate method is to **decrease the threshold of classifying a point to be positive properly.**

$W_n = 1, W_p = 50$	
Performance Measures	Test Performance
Accuracy	0.3125
Precision	0.1697
F1-Score	0.2894
AUROC	0.8158
Average Precision	0.4833
Sensitivity	0.9825
Specificity	0.2012

Table 4: Performance with arbitrary class weights

Figure 5: Table of 3.1(b).

4 Kernels

4.1 Comparing Ridge and Logistic Regression

- (a)
 - **Question:** Write out Ridge Regression’s objective function in terms of $\bar{\theta}$ (in this case the normal objective function instead of the kernelized function). How does it differ from the objective function associated with l_2 -regularized logistic regression?

– **Answer:**

$$J(\bar{\theta}) = \lambda \frac{\|\bar{\theta}\|^2}{2} + \frac{1}{2n} \sum_{i=1}^n (y^{(i)} - \bar{\theta} \cdot \bar{x}^{(i)})^2 \quad (1)$$

Here, it uses **squared loss** other than logistic loss.

- (b)
 - **Question:** Report the table. Do you observe a large difference in performance between the two approaches? Why or why not?
 - **Answer:** See Figure 8. I don’t observe a large difference. The reason behind is maybe because they both are **linear models**.

$W_n = 1, W_p = 4$	
Performance Measures	Test Performance
Accuracy	0.8000
Precision	0.3736
F1-Score	0.4595
AUROC	0.8207
Average Precision	0.4916
Sensitivity	0.5965
Specificity	0.8338

Table 5: Performance with chosen class weights

Figure 6: Table of 3.2(b).

4.2 RBF Kernel

- (b)
 - **Question:** Report in Table 4.2(b) and explain the effect γ has on cross-validation performance.
 - **Answer:** See Figure 9. Increasing γ will first enhance the CV performance. After reaching a peak between 1 to 10 in this case, it will decrease the performance.
- (c)
 - **Question:** Report the test performance of this model on the test data in Table 4.2c.
 - **Answer:** See Figure 10.
- (d)
 - **Question:** Is it possible to report the weight associated with each feature in the final model? Why or why not?
 - **Answer:** No, it is impossible to report each coefficients. It is because the RBF kernel will deal with the combination of these features when classifying. Those high dimensional variables are not easily split back to the single features.

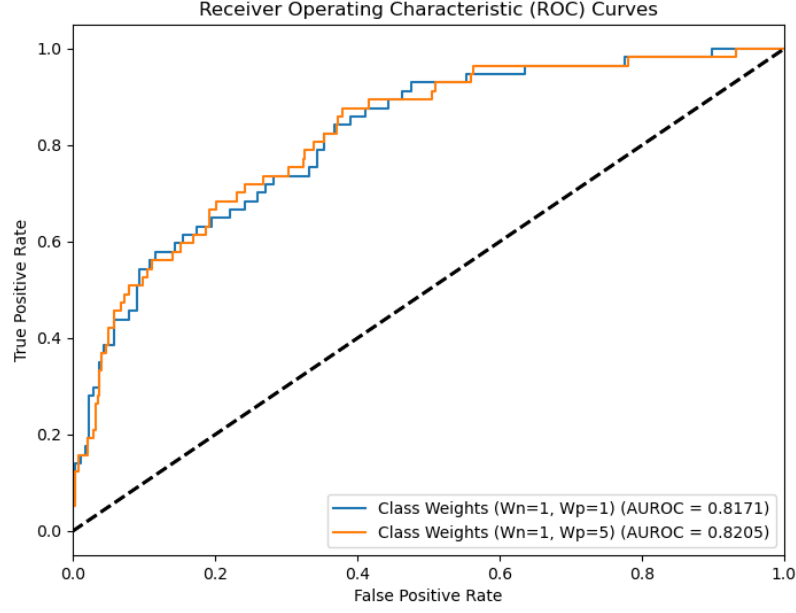


Figure 7: Table of 3.3(a).

$C = 1.0$, penalty = ℓ_2		
Metric	ℓ_2 -Regularized Log. Regression	Ridge Regression
Accuracy	0.8675	0.8725
Precision	0.6429	0.7500
F1-Score	0.2535	0.2609
AUROC	0.8172	0.5746
Average Precision	0.4985	0.2384
Sensitivity	0.1579	0.1579
Specificity	0.9854	0.9913

Table 6: Test Performance of ℓ_2 -Regularized Log. Regression and Ridge Regression

Figure 8: Table of 4.1(b).

γ	Mean (Min, Max) CV Performance
0.001	0.7749 (0.7474, 0.8028)
0.01	0.7929 (0.7572, 0.8433)
0.1	0.8034 (0.7634, 0.8631)
1.0	0.8036 (0.7594, 0.8688)
10	0.7715 (0.7410, 0.8225)
100	0.7428 (0.7146, 0.7802)

Table 7: Cross-validation Performance with different γ

Figure 9: Table of 4.2(b).

$C=0.1, \gamma=1$	
Performance Measures	Test Performance
Accuracy	0.8700
Precision	0.7778
F1-Score	0.2121
AUROC	0.8273
Average Precision	0.5134
Sensitivity	0.1228
Specificity	0.9942

Table 8: Test Performance of KernelRidge Model with C and γ that maximize AUROC

Figure 10: Table of 4.2(c).

5 Challenge

5.1 Performance Metrics

In this Challenge, I employed the average of AUROC(mean) and F1-Score(mean) as the performance metrics, indicating predictive power.

5.2 Feature Extraction

5.2.1 Representative Statistics

- Separate the static and time-varying variables apart, and choose some informative representative statistics for time-varying variables.

- I added the minimum value and standard derivation because some indicators' minimum may indicate a dangerous state for the patient; some indicator's variance during the 48 hours may remain in a stable interval to keep healthy.

5.2.2 Small Windows

- Split the 48 hours into smaller periods.
- Since in one day, some features may vary regularly. I decided to not split further, but just split into two windows. Each one is a day. For each window, calculate the maximum, minimum, and standard derivation as the representative statistics.

5.2.3 Missing Value

- Try other way of choosing median to be the missing value.
- Though median and mean have different meanings, in this example, the results are quite similar. There's not a specific reason to choose one. However, based on the performance measure in general, mean may be a valid choice.

5.2.4 Conclusion

In Feature Extraction, I further incorporate these methods. For 5.2.3, I didn't apply it finally since it does not help to improve the performance and is not supported by reasoning. For 5.2.2, I decided to keep the **maximum and minimum of each period**, but use **the standard derivation of the whole 48 hours**. That's because the variation of 48 hours will give us more information and the variances of each period seem to be independent (meaning we don't have a proper reason to split into them). On the other hand, the performance in general is optimized in this way.

5.3 Hyperparameter Selection

5.3.1 Class Weight

I used 1:4 as the final class weight since it optimizes the performance among every test cases compared to 1:1, 1:2, 1:3, 1:5, ..., 1:50.

5.3.2 Penalty Norm & C_range

With Class Weight fixed, I tried through two penalty norms ("l1", "l2"), 6 C_ranges (0.001, 0.1, 1, 10, 100). The ("l1", 1) pair is the best optimizer.

5.3.3 Further Tuning & Conclusion

I then used **Bisection method** to further tune C starting [0.8, 1.2]. Finally, the optimizing solution is around 0.85. Thus, the hyperparameters are

$$W_n = 1, W_p = 4, C = 0.85, norm = "l1". \quad (2)$$

5.4 Result

The AUROC and F1-score are:

$$AUROC = 0.8342, F1_Score = 0.5116. \quad (3)$$

The confusion matrix applied to $X_challenge$ and $y_challenge$ is shown below.

$$\begin{bmatrix} 7055 & 1405 \\ 472 & 1068 \end{bmatrix}$$

That is "*True Negatives (7055); False Positives (1405); False Negatives (472); True Positives (1068)*".